Universal Plastic Shredder V2.0

Electrical Control and Power System

Final Report - Rough Draft

Team 07: I'm So Shredded INC
Bao Nguyen, Fearghus Tyler, Yaqoub Rabiah, Fox Kang
baon@pdx.edu, fearghus@pdx.edu, yrabiah@pdx.edu, foxkang@pdx.edu

Industry Sponsor: MME / Ichor Systems (Bridget Lannigan)
Faculty Advisors: Prof. Andrew Greenberg, Dr. Jonathan Bird (EE), Robert Paxton (ME)

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[NOTE: This is a rough draft. Sections marked TODO, NOTE, or QUESTION indicate areas where we need to expand content, gather data, or get advisor input before the final submission. Bullet points are placeholders for fuller prose in the final version.]

Contents

1 Executive Summary

- Ichor Systems generates significant plastic waste (PVC, polypropylene, LDPE) during semiconductor equipment manufacturing, currently sent to landfill.
- Commercial industrial shredders cost $10,000+ and consumer units cannot handle the thick rigid sheets (0.25–0.5 in).
- Our project designs the electrical control and power system for a custom industrial-grade plastic shredder, capable of shredding into 1-inch pieces.
- Key elements: VFD-controlled 3 HP motor at 70–90 RPM output, two-hand safety control, lid interlock, E-stop, jam-detection auto-recovery with 3-strike lockout, NEMA-enclosed PLC and HMI.
- Total budget: $3,000 (EE portion approximately $1,000).

[QUESTION: Should the executive summary include high-level results / status (e.g., "shredder validated to X HP load with Y throughput rate") once we complete testing? Currently we have signal-chain validation but not full mechanical load testing.]

2 Introduction

2.1 Background and Motivation

- Ichor Systems is a Portland-area semiconductor equipment manufacturer.
- Their manufacturing processes produce substantial volumes of rigid plastic sheet waste.
- Existing disposal route: landfill – environmentally and economically suboptimal.
- A previous 2023–2024 PSU capstone team built a 2 HP household shredder for the EPL. This project scales to industrial-grade.

2.2 Project Scope

- EE team responsibility: motor control, safety circuits, PLC logic, HMI, power distribution.
- ME team (separate) handles mechanical frame, blade assembly, gearbox.
- Integration plan: mechanical structure delivered with motor mount; EE team commissions controls.

[TODO: Add a one-paragraph background on dynamic recycling / circular economy framing for the introduction.]

3 Requirements

3.1 Must-Have

- Shred 0.25–0.5 in plastic sheet into approximately 1-inch pieces.
- Controllable VFD supporting 3–5 HP motor at 70–90 RPM output shaft speed.
- Follow NFPA 70 (NEC) electrical design guidance.

- Include emergency stop, overcurrent/overload protection, at least one safety interlock.
- Include an HMI for operator control and status display.
- Stay within $3,000 total project budget.

3.2 Should-Have

- Clear status and fault indicator lights.
- Follow NEMA enclosure guidelines.
- Clearly labeled controls.

3.3 May-Have (if time/budget allow)

- Wireless monitoring (read-only).
- Data logging of operating hours and faults.
- Additional safety sensing (capacitive touch, thermal).

4 System Architecture

4.1 Hardware Subsystems

- **AC power input:** 120 VAC mains → main disconnect → branch breakers → VFD and 24V PSU.
- **VFD:** Huanyang HY02D211B-T (2.2 kW, 110V single-phase input, 220V 3-phase output boost-type). Controlled via FOR/REV/RST digital inputs.
- **Motor:** GE 5KE182BC205B, 3 HP, 230V (low-voltage connection), 1750 RPM, FLA 7.6 A.
- **Brake resistor:** 75 Ω, 500 W, connected to VFD's P/Pr terminals to handle regenerative energy during deceleration and reversal.
- **PLC:** AutomationDirect H2-DM1E (Do-more CPU) on DL205 base, D2-08ND3 discrete input module.
- **HMI:** AutomationDirect C-more Micro-Graphic EA1-T4CL 4-inch color touch panel, connected to the H2-DM1E CPU serial communication port (operator commands to the PLC; status/data back to the panel). Powered from the shared Mean Well NDR-480-24 24 VDC supply.
- **Safety circuit:** Hardwired E-stop loop, lid interlock, two-hand deadman start. Motor contactor energized through safety chain.
- **24V DC supply:** Mean Well NDR-480-24 (480 W) for control voltage.

4.2 Signal Flow

- Operator inputs (deadman buttons, E-stop, reset, lid interlock) → PLC discrete inputs (D2-08ND3, sourcing mode).

- PLC outputs (D2-08TR relay module, common to PSU –V) → VFD FOR (Y7), REV (Y6), RST (Y5) inputs in sinking (NPN) mode.
- VFD fault relay (FC + FA, NO active-high) → PLC X5 input. X5 = HIGH when VFD trips on fault, LOW when healthy.
- HMI ↔ PLC via serial (H2-DM1E CPU serial COM port; matched baud, parity, and stop bits on both ends).

[TODO: Insert system block diagram here. Convert one of the AutoCAD drawings or draw fresh in draw.io.]

5 Hardware Design

5.1 Power Distribution

- Main disconnect: **[TODO: specify rating]**.
- VFD branch breaker: 30 A C-curve DIN-rail (handles inrush, sized per NEC 430.52).
- 24V PSU branch breaker: 10 A C-curve DIN-rail.
- **Conductors:** power runs use 12 AWG copper (THHN/MTW), rated 20 A / 25 A / 30 A at the 60/75/90 °C ampacity columns of NEC Table 310.16 – well above the motor's 7.6 A FLA and the VFD input current. Control wiring uses 16–18 AWG, ferruled at all terminal blocks.
- **Fuse / OCPD sizing:** branch and component fuses are sized at **125% of the protected component's rated continuous load** per NEC 210.20(A) / 215.3. (The motor branch breaker is the exception – NEC 430.52 permits a higher multiple of FLC to ride through inrush; running overload is handled by the VFD's electronic motor-overload function.)
- All branches fused individually at terminal blocks for downstream loads.

5.2 Motor and VFD

- Selected Huanyang HY02D211B-T over alternatives via decision matrix (price, documentation, reliability).
- Programmed motor parameters: `PD141 = 230 V`, `PD142 = 7.6 A`, `PD143 = 4 poles`, `PD144 = 1750 RPM`.
- Over-torque detection programmed: `PD052 = 02` (fault indication on relay), `PD123 = 3`, `PD124 = 150` (%), `PD125 = 3.0 s`.
- Brake resistor connected to P/Pr terminals to handle regenerative energy from rapid reversal during jam-clear cycles.

[NOTE: The over-torque relay function (`PD052 = 12`) did not fire the relay in our specific Huanyang revision when `PD123 = 2` (continue running). Documented as a firmware-specific limitation (see the VFD parameter table, Appendix D). Working configuration uses `PD052 = 02` (fault indication) and `PD123 = 3` (over-torque triggers self-trip, which fires the fault relay). The PLC then runs the unjam routine including a RST pulse to clear the VFD's latched fault state between retry attempts.]

5.3 Safety Circuit

- E-stop: NC contact pair in hardwired loop. Drops motor contactor on activation.
- Lid interlock: NC switch in same hardwired safety loop.
- Two-hand deadman: both buttons must be held continuously to enable motor.
- Motor contactor: K1 main contactor energized through safety chain. PLC outputs RUN command *through* the contactor, so safety chain has hardware precedence over PLC.

[TODO: Add safety circuit schematic. Reference NFPA 79 section 9 for safety-related stop functions.]

5.4 Jam Detection Signal Chain

- VFD's multi-function relay (terminals FA, FB, FC) programmed for Fault Indication (`PD052 = 02`). **Only FA and FC are used** for jam detection; the FB (normally-closed) contact is left unconnected.
- Wired as an NO active-high pair: +24V (through a 500 mA fast-blow fuse) feeds the FA contact; the FC common returns to PLC X5 input on D2-08ND3 (Slot 2). When the VFD trips, FA–FC closes and +24V reaches X5.
- PLC sees X5 = LOW during normal operation, X5 = HIGH when VFD trips on any fault (over-torque, overvoltage, overheat, etc.).
- PLC ladder reads `[ X5 ]` on rising edge to detect new fault events.
- Wire-break failure mode is handled at the safety-circuit level by the redundant hardwired E-stop loop and motor contactor, rather than at the jam-signal level.
- Full terminal-by-terminal wiring is given in the I/O point list (Appendix B) and the VFD parameter/wiring table (Appendix D).

6 Software Design

6.1 PLC Ladder Logic Architecture

- Do-more Designer stage program (similar to grafcet / SFC).
- States: IDLE, RUNNING, JAM_DETECTED, RST_PULSE, REVERSE, RESUME, LOCKOUT.
- Transitions driven by deadman input, X5 jam signal, jam counter, and reset button.

6.2 State Machine

- **IDLE** → RUNNING when both deadman buttons held AND not locked out AND startup mask done.
- **RUNNING** → JAM_DETECTED on rising edge of X5 (VFD trip → fault relay fires → X5 goes HIGH).
- **JAM_DETECTED** → RST_PULSE if jam count < 3, → LOCKOUT if jam count ≥ 3.

- **RST_PULSE** (500 ms) → REVERSE (2 s).
- **REVERSE** → RESUME (brief delay) → back to RUNNING.
- **LOCKOUT** → IDLE only via dedicated reset button.

6.3 Jam Counter Reset Logic

- Counter increments on each X5 rising edge (each new VFD fault triggers a jam event).
- Resets to 0 if motor runs cleanly for 30 continuous seconds in the RUNNING state.
- Also resets if deadman is released for 30 continuous seconds (session reset).
- Whichever condition fires first wins.

6.4 Startup Mask

- For first 5 seconds after PLC power-up, X5 is ignored.
- In NO active-high wiring, this guards against transient X5 pulses during VFD initialization. Less critical than under NC wiring but kept as a defensive measure.

6.5 HMI Programming

- Status displays: Ready, Running, Jam in progress, Lockout, Fault.
- Operator inputs: Speed setpoint (currently fixed in PLC), Reset button.
- Data displayed: Motor current (read from VFD over Modbus – **[TODO: verify Modbus setup]**), Jam count this session, Total runtime.

[QUESTION: We have not finalized the HMI screen layout. Should we include mock-ups in the rough draft, or wait for the final?]

7 Safety Design

7.1 NFPA 79 Compliance

- Wire color coding follows NFPA 79 §13.2: Black for AC line/load, Red for AC control, Blue for DC control, Blue/W stripe for DC return, Green/Yellow for ground.
- This color convention is applied to every conductor in the panel.
- All control wires labeled at both ends.
- All control wires terminated with crimped ferrules at terminal blocks.

7.2 Hardwired Safety Functions

- E-stop: Category 0 stop (immediate removal of power) per ISO 13849. NC fail-safe wiring.
- Lid interlock: NC fail-safe; opens motor contactor when lid raised.
- Two-hand control: per ISO 13851 (Type IIIA at minimum) – buttons must be pressed within 0.5 s of each other and held continuously.

- Overload: motor thermal protection in VFD plus separate motor contactor with overload relay.

[TODO: Confirm we are meeting Category 1 or higher per ISO 13849-1. Need to walk through the SIL/PL calculation or get a hand-wave from the advisor that capstone scope doesn't require formal SIL.]

7.3 Failure Mode Analysis

- Wire break in jam signal: PLC sees X5 stuck LOW. Jam signal cannot fire, but the hardwired E-stop loop and motor contactor provide independent safety. Operator can also see motor stop and trigger investigation via fault-light absence.
- VFD loses power: the FA–FC fault relay de-energizes and opens, so X5 reads LOW (indistinguishable from healthy). The motor stops because the VFD itself is unpowered, so the condition is caught by the motor not running rather than by the jam signal.
- PLC fails: motor contactor drops out (safety chain bypasses PLC). Motor coasts to stop.
- Brake resistor disconnected: VFD will trip on overvoltage during deceleration. Motor coasts to stop. Diagnosable from VFD display.

8 Standards and Regulatory Compliance

This section consolidates every electrical and machine-safety standard the design is built against and states, for each, how it is applied to the shredder. Many of these are referenced in passing elsewhere in this report; they are gathered here so a reviewer (or a future maintenance team) can audit compliance in one place.

8.1 Standards Summary Matrix

Standard	Scope	How it is applied here
NFPA 70 (NEC)	National Electrical Code – premises wiring, motor circuits	Branch-circuit, conductor, grounding, and disconnect design (see §8.2).
NFPA 79	Electrical standard for industrial machinery	Wire color coding, supply disconnect, protective bonding, stop functions, operator devices (see §8.3).
NEMA 250	Enclosure type ratings	Control enclosure selected to a NEMA type matched to the lab/shop environment (see §8.4).
NEMA ICS 1/2	Industrial control & systems	Contactor / control device selection and marking.
NEMA MG 1	Motors and generators	Motor nameplate interpretation and VFD

8.2 NFPA 70 – National Electrical Code

- **Article 430 (Motors, Motor Circuits, and Controllers)** – the governing article for this build.
  - `430.6`: motor full-load current taken from the nameplate / NEC tables (GE 5KE182BC205B, FLA 7.6 A at 230 V) as the basis for all downstream sizing.
  - `430.22`: branch-circuit conductors sized at $\geq 125\%$ of motor FLC.
  - `430.32`: running overload protection – provided by the VFD's electronic motor-overload function (programmed motor FLA `PD142 = 7.6 A`) plus over-torque trip.
  - `430.52`: branch-circuit short-circuit / ground-fault protection – 30 A C-curve breaker on the VFD branch, sized to clear inrush without nuisance trips (within the inverse-time-breaker allowance).
  - `430.102` / `430.109`: a disconnecting means is provided ahead of the controller and VFD.
  - **Part X (Adjustable-Speed Drive Systems, `430.122`)**: conductor and protection rules applied to the VFD-fed motor circuit.
- **Article 250 (Grounding and Bonding):** equipment grounding conductor (green / green-yellow) bonds the VFD chassis, PSU chassis, motor frame, and panel to a common ground bus and building earth.
- **Article 310 (Conductor Ampacity):** control and power conductors selected from the ampacity tables for the installed temperature rating.
- **Article 409 (Industrial Control Panels):** panel layout, marking, and overcurrent considerations.
- **Article 110.26 (Working Space):** clearances around live parts maintained for safe service access. **[TODO: Confirm final panel mounting leaves required working clearance.]**

[NOTE: As-built deviation flagged during the Point List review: the SD-N35 safety contactor switches the VFD neutral *(T) leg rather than the* hot *(R) leg. NEC / NFPA 79 practice is to switch the ungrounded (hot) conductor so that opening the contactor de-energizes the load to ground potential. Switching the grounded conductor leaves the load at line potential when the contactor is open. This is documented in the as-built I/O point list (Appendix B, VFD terminals) and must be corrected, or explicitly justified and risk-assessed, before final sign-off.]*

8.3 NFPA 79 – Electrical Standard for Industrial Machinery

- **§5.3 Supply circuit disconnecting means:** a lockable main disconnect isolates the machine from the supply (also satisfies LOTO, below).
- **§7 Overcurrent protection:** individual branch and downstream-load fusing at the terminal blocks (PSU branch 25 A slow-blow, PLC branch 1 A fast-blow, control loads individually fused).
- **§8 Protective bonding circuit:** continuous equipment-grounding/bonding path to all exposed conductive parts.

- **§9 Control circuits and functions:** stop functions classified by stop category (see safety section); the hardwired safety chain has precedence over the PLC.
- **§10.7–10.8 Operator interface / emergency stop:** E-stop uses a red mushroom actuator on a yellow background with a self-latching NC contact; start devices are guarded against accidental operation by the two-hand requirement.
- **§13.2 Conductor identification (color coding):** black = AC line/load, red = AC control, blue = DC control, blue/white = grounded (return) DC, green/yellow = protective earth. All control wires are labeled at both ends and terminated with crimped ferrules.

8.4 NEMA Standards

- **NEMA 250 (Enclosures):** the control enclosure is selected to a NEMA type appropriate for a dry indoor shop/lab with airborne plastic dust – targeting NEMA Type 12 (dust-tight, drip-tight) for the PLC/HMI panel. **[TODO: Confirm the final enclosure's NEMA type rating and that dust ingress from shredding is addressed.]**
- **NEMA ICS 1 / ICS 2 (Industrial Control & Systems):** contactor and control-device selection and marking follow NEMA control conventions; the Mitsubishi SD-N35 contactor is sized above the motor FLA for the AC-3 (motor) duty.
- **NEMA MG 1 (Motors and Generators):** used to interpret the GE motor nameplate (voltage connection, poles, RPM, service factor) when programming the VFD motor parameters (`PD141--PD144`).

8.5 Machine-Safety Standards (ISO / IEC)

- **ISO 12100:** risk-assessment framework that identifies the shredder's primary hazards (rotating blades, entanglement, electrical) and drives the layered safeguards.
- **ISO 13849-1:** the E-stop / lid-interlock / two-hand chain targets a performance level (PL) commensurate with the assessed risk. **[TODO: Walk through the PL/category calculation or obtain advisor confirmation that formal SIL/PL verification is out of capstone scope.]**
- **ISO 13850:** E-stop implemented as a Category 0 stop (immediate removal of power) with fail-safe NC wiring.
- **ISO 13851:** two-hand deadman control – both buttons must be actuated within 0.5 s of one another and held continuously to enable the motor.
- **IEC 60204-1:** the international analogue of NFPA 79; used to cross-check stop-category definitions and supply-disconnection requirements.

8.6 Occupational Safety (OSHA)

- **29 CFR 1910.147 (Lockout/Tagout):** the main supply disconnect is lockable so the machine can be positively isolated for blade-area service and maintenance.
- **29 CFR 1910 Subpart S (Electrical):** general electrical-safety practices (guarding of live parts, grounding) observed throughout the panel build.

[TODO: Before final submission: convert any remaining "targets / intends to" language above into verified compliance statements, attach the panel enclosure datasheet, and resolve the NFPA 79 neutral-switching deviation note.]

9 Testing and Validation

9.1 Signal Chain Tests

- Manual jumper test (FA-FC short): confirmed PLC X5 reads LOW correctly – wiring intact.
- Guaranteed-trip test (`PD124 = 1`, `PD125 = 5.0`): VFD trips reliably on motor run, fault relay fires, PLC X5 drops.
- Threshold tuning: 150% (`PD124 = 150`) chosen for production based on no-load current (0.5 A) versus expected jam current (above 11 A actual).

9.2 Load Testing

[TODO: Schedule load testing with actual plastic samples once mechanical integration complete. Need ME team coordination.]

9.3 Safety Validation

- E-stop response time: **[TODO: measure with scope]**.
- Lid interlock response: **[TODO: measure]**.
- Two-hand control timing: **[TODO: verify both buttons must be held continuously, simultaneous release stops within X ms]**.

10 Results

10.1 What Works

- All motor parameters programmed and verified.
- Over-torque detection chain validated (VFD detects, relay fires, PLC reads, ladder reacts).
- Fail-safe NC wiring confirmed – system correctly enters fault state on signal chain failure.
- HMI communicates with PLC over Ethernet.

10.2 Open Issues

- Vendor-specific Huanyang firmware quirk: `PD052 = 12` does not fire relay in continue-running mode. Working around with `PD052 = 02` + `PD123 = 3`.
- PLC ladder logic not fully complete – state machine implemented, but unjam routine timing needs tuning with real loaded motor.
- Mechanical integration pending – have not yet tested under real shredding load.

[QUESTION: How much detail should we include about the Huanyang firmware quirk? It is an interesting design lesson (vendor lock-in, never trust the datasheet completely) but distracts from the high-level results.]

11 Discussion

11.1 Design Tradeoffs

- Cost versus reliability: chose Huanyang VFD ($150) over ATO ($450). Firmware quirks documented in this report (the Software Design note and Appendix D) for future maintenance.
- 110V single-phase versus 220V split-phase mains: 110V chosen for receptacle availability in PSU lab; trade-off is double the input current and need for a dedicated 30 A circuit.
- NC fail-safe versus NO active-high jam signal: chose NC for safety; trade-off is potential nuisance lockouts from broken wires (acceptable for shredder).

11.2 Lessons Learned

- Always verify VFD relay function with multimeter before assuming firmware works as documented.
- Document deviations from datasheet behavior in version control so future teams can find them.
- Power cycle after parameter changes – some VFDs require it for save to fully take effect.
- Use ferrules on all control wires – loose strand connections caused multiple debugging hours during commissioning.

11.3 Project Management Notes

[TODO: Add a paragraph on how the team coordinated, what tools we used (GitHub, weekly meetings, etc.), and how schedule slipped or held.]

12 Future Work

- Wireless monitoring (read-only) – could expose PLC data via the H2-DM1E's built-in web server or an ESP32 bridge.
- Data logging: total runtime, jam events per shift, faults. Useful for predictive maintenance.
- Predictive jam detection: train a small model on current waveform shape just before jams to detect them earlier.
- Mechanical integration with feed hopper that auto-meters plastic into the blade.
- Higher-capacity model: scale to 5–10 HP for higher throughput.

13 Ethics and Design Considerations

A summary of the ethics and design evaluation follows:

- **Public health and safety:** Yes – multiple redundant safety interlocks, NFPA 70 / NEMA compliance, hardwired safety chain.
- **Environmental:** Yes – core project purpose is diverting plastic from landfill.

- **Economic:** Yes – significant cost savings versus commercial alternatives, careful component selection within budget.
- **Global:** Possibly – approach is replicable globally, components internationally sourced.
- **Cultural:** No – industrial machine for B2B use, no culture-specific design.

14 References

- NFPA 70 (2023): National Electrical Code (esp. Articles 110, 250, 310, 409, 430).
- NFPA 79 (2024): Electrical Standard for Industrial Machinery.
- NEMA 250: Enclosures for Electrical Equipment.
- NEMA ICS 1 / ICS 2: Industrial Control and Systems.
- NEMA MG 1: Motors and Generators.
- UL 508A: Industrial Control Panels.
- IEC 60204-1:2016: Safety of Machinery – Electrical Equipment of Machines.
- ISO 12100:2010: Safety of Machinery – Risk Assessment and Risk Reduction.
- ISO 13849-1:2023: Safety-related parts of control systems.
- ISO 13850:2015: Emergency Stop Function.
- ISO 13851:2019: Two-hand control devices.
- OSHA 29 CFR 1910.147: The Control of Hazardous Energy (Lockout/Tagout).
- Huanyang HY-series VFD User Manual.
- AutomationDirect DL205 / Do-more User Manual.
- GE 5KE Frame 182T Motor Datasheet.

[TODO: Add full reference list with URLs / publication info before final.]

15 Appendices

15.1 Appendix A: Bill of Materials

Major purchased items for the electrical system (procurement BOM, rev 02/20/26):

Item	Description	Mfr. P/N	Critical
Motor	AC induction motor, 145TC frame, 3-phase TEFC	PE145TC-2-4	Yes
VFD	Huanyang 1-phase in / 3-phase out drive, 3 HP	HY02D211B-T	Yes
HMI	C-More Micro-Graphic 4” color touch panel	EA1-T4CL	Yes
Circuit breaker	DIN-rail branch breaker	NBE7	No
Power switch	Single-phase ON/OFF disconnect, 120–230 V	71008	No
Status lights	Red / green panel indicators	Ichor-provided	No
E-stop	Mushroom emergency-stop pushbutton	Ichor-provided	Yes

*[NOTE: The procurement BOM lists motor **PE145TC-2-4**; the as-built control design in this report is parameterized for a GE 5KE182BC205B (3 HP, 230 V, 7.6 A, 4-pole, 1750 RPM). Reconcile the motor line item against the installed motor before final. Additional control-panel hardware (PLC rack/modules, Mean Well NDR-480-24 PSU, SD-N35 contactor, Phoenix Contact breaker, brake resistor, illuminated pushbuttons, fuses) is itemized in the component tables of the Hardware Design section and the point list in Appendix B.]*

15.2 Appendix B: PLC I/O Point List

As-built I/O assignments for the DirectLOGIC 205 (D2-09B-1) base.

Rack slot layout

Slot	Module	Function
1	H2-DM1E	CPU with built-in Ethernet; serial COM port to HMI
2	D2-08ND3	8-ch discrete input, 24 VDC sourcing (X0–X7)
3	F2-08AD-1	8-ch analog current input; CH1 = VFD VO motor-current monitor
4	F2-04RTD	4-ch RTD input (all spare)
5	D2-08TR	8-ch relay output, sinking, common to PSU −V (Y0–Y7)
6	F2-08DA-2	8-ch analog voltage output; CH1 (+V1) = VFD VI speed reference
7–9	D2-Fill	Fill panels (reserved)

Discrete inputs (D2-08ND3, sourcing; common to PSU −V)

Addr	Field device / function
X0	Forward pushbutton (Autonics S2PR
X1	Reverse pushbutton (S2PR-P3Y, NO
X2	Run 1 / Deadman 1 (S2PR-P3B, NC
X3	Run 2 / Deadman 2 (S2PR-P3B, NC
X4	E-stop monitor tap (E22B1 NC, in se
X5	VFD fault relay (FA–FC, NO active-
X6	Reserved – Reset pushbutton (S2PR-
X7	Spare

Relay outputs (D2-08TR, sinking; common to PSU −V)

Addr	Load
Y0	Forward indicator LED (SA-LDG green)
Y1	Reverse indicator LED (SA-LDY yellow)
Y2	Run 1 / Deadman 1 LED (SA-LDB blue)
Y3	Run 2 / Deadman 2 LED (SA-LDB blue)
Y4	Error / fault indicator (Wamco 525, 28 V
Y5	VFD RST (fault-reset pulse, ∼500 ms)
Y6	VFD FOR (forward run command)
Y7	VFD REV (reverse run command)

Analog and communications

Point	Connection
F2-08AD-1 CH1+	VFD VO terminal – motor-current monitor (0–10 V, PD054
F2-08DA-2 +V1	VFD VI terminal – speed reference (0–10 V, PD002=1)
CPU serial COM	EA1-T4CL HMI (operator commands / status & data)
HMI power	Mean Well NDR-480-24 (+V / −V), 24 VDC

VFD terminals (Huanyang HY02D211B-T)

Terminal	Connection
R	AC hot via main disconnect + 30 A C-cu
T	AC neutral via Phoenix Contact TMC 8 3(pole (as-built: neutral switched – see NEC
U / V / W	Motor T1/T2/T3 via colored banana sock
P+ / PR	Brake resistor (ATO APCS-300R30-AD, 3
VI / VO	PLC F2-08DA-2 +V1 / PLC F2-08AD-1
FOR / REV / RST	PLC D2-08TR Y6 / Y7 / Y5 (sinking, pu
FA / FC	PSU +V / PLC X5 (fault relay, PD052=(
DCM / ACM	PSU −V (shared digital / analog 0 V refe

15.3 Appendix C: Wiring Diagrams

[TODO: Insert exported PDFs from AutoCAD: power, VFD/motor, PLC/HMI, power distribution.]

15.4 Appendix D: VFD Parameter Settings

Programmed parameters on the Huanyang HY02D211B-T (as-built):

Param	Value	Meaning
PD002	1	Frequency source = external analog voltage (VI, 0–10 V from PLC)
PD052	02	Multi-function relay (FA/FC) = fault indication
PD054	1	Analog output VO = output current (0–10 V monitor to PLC)
PD123	3	Over-torque action = trip/fault (fires the fault relay)
PD124	150	Over-torque detection threshold (% of rated)
PD125	3.0	Over-torque detection time (s)
PD141	230	Motor rated voltage (V)
PD142	7.6	Motor rated current (A)
PD143	4	Motor poles
PD144	1750	Motor rated speed (RPM)

Power / braking wiring: R = AC hot (30 A breaker); T = AC neutral via SD-N35 contactor; U/V/W to motor via colored banana sockets; P+/PR to a 300 W, 30 Ω brake resistor (ATO APCS-300R30-AD) to absorb regenerative energy during reversal in the jam-clear cycle. Control wiring is summarized in Appendix B.

[NOTE: Firmware-specific finding: `PD052 = 12` (continue-running over-torque indication) did not fire the relay on our Huanyang revision; the working configuration is `PD052 = 02` with `PD123 = 3` (over-torque self-trips, which fires the fault relay and is read by the PLC on X5).]

15.5 Appendix E: Test Plan

Test Plan v1.0 comprises a top-down power-on walkthrough followed by ten bottom-up test cases. Each case is exercised against the must/should requirements (M-series).

ID	Test case
Top-down	Power-on walkthrough: AC in → PSU → PLC boot → HMI IDLE/READY
BU-01	AC power input and fuse distribution
BU-02	24 VDC PSU output (Mean Well NDR-480-24)
BU-03	PLC rack initialization (DirectLOGIC 205)
BU-04	E-stop hardware safety circuit
BU-05	Two-hand start (deadman) safety logic
BU-06	Lid interlock safety input
BU-07	VFD speed command and motor RPM (HY02D211B-T)
BU-08	Overcurrent / overload protection
BU-09	HMI operator interface (EA1-T4CL C-More Micro-Graphic)
BU-10	Status indicator lights (power / running / fault)

Validation results to date are summarized in the Testing and Validation and Results sections; load testing under real shredding load is pending mechanical integration.

End of rough draft. Comments and questions welcome.